Falsification of $R(8,8) > 293$: Exhaustive Max-Clique Verification and the Structural Limits of Stochastic Sampling

Bryan Daugherty[1], Gregory Ward[1], Shawn Ryan[1]

[1]SmartLedger Solutions / Origin Neural AI

bryan@smartledger.solutions greg@smartledger.solutions
shawn@smartledger.solutions

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Abstract

We report the falsification of the claim $R(8,8) > 293$ through exhaustive max-clique verification of a GPU-optimized two-coloring of the complete graph K_{293}. A Bron–Kerbosch search with Tomita pivoting reveals a monochromatic K_8 in the red subgraph at vertices $\{3, 44, 87, 130, 165, 219, 234, 285\}$ and a monochromatic K_{11} in the blue subgraph, despite GPU-accelerated stochastic sampling (1.5 million random 8-clique tests across three seeds) reporting zero estimated violations. This establishes a concrete failure mode of sparse stochastic verification at the scale of $\binom{293}{8} \approx 2.26 \times 10^{14}$ candidate cliques, where random sampling covers only $\sim 6.6 \times 10^{-9}$ of the search space. We confirm $R(8,8) > 281$ via a Paley construction proof: the quadratic-residue coloring of K_{281} is certifiably K_8-free in both colors. Additionally, we verify the Zero-Core Theorem for $R(5,5)$: all 2,480 monochromatic K_4 constraints in the champion K_{43} coloring's K_{44} extension are non-essential, proving the structural obstruction is fully distributed with an empty essential core. All results are computationally reproducible with zero external dependencies.

Keywords: Ramsey theory, diagonal Ramsey numbers, graph coloring, max-clique, Bron–Kerbosch, Paley construction, stochastic verification, GPU optimization, Zero-Core Theorem

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Table 1: Summary of principal results and epistemic status.

Result	Value	Status
$R(8,8) > 293$ claim	**Falsified**	**[Proved]** Exhaustive verification
Red max-clique in K_{293}	$\omega = 8$ (monochromatic K_8)	**[Proved]** Bron–Kerbosch
Blue max-clique in K_{293}	$\omega = 11$ (monochromatic K_{11})	**[Proved]** Bron–Kerbosch
Stochastic sampling coverage	6.6×10^{-9} of $\binom{293}{8}$	**[Computational]** Measured
$R(8,8) > 281$ (Paley)	**Confirmed**	**[Proved]** Exhaustive K_8 enumeratic
Paley(293) violations	2,310,012 monochromatic K_8	**[Computational]** 31.1 s
Zero-Core Theorem ($R(5,5)$)	Essential core = $\varnothing$	**[Proved]** 2,480 DPLL proofs

1 Introduction

1.1 Historical Context

The diagonal Ramsey number $R(k,k)$ is the smallest integer n such that every two-coloring of the edges of the complete graph K_n contains a monochromatic complete subgraph K_k. Since Ramsey's foundational theorem [1], determining exact values of $R(k,k)$ has remained one of the most challenging problems in combinatorics.

For small values, exact results are known:

Table 2: Known diagonal Ramsey numbers. Best bounds for $k \geq 5$ as of 2025.

k	$R(k,k)$	**Status**	**Reference**
1	1	Exact	Trivial
2	2	Exact	Trivial
3	6	Exact	Ramsey [1]
4	18	Exact	Greenwood & Gleason [4]
5	$43 \leq R(5,5) \leq 48$	Bounds	DWR [14] (lower); McKay & Radziszowski [10] (upper)
6	$102 \leq R(6,6) \leq 165$	Bounds	Exoo [8]; Bohman & Keevash [11]
7	$205 \leq R(7,7) \leq 298$	Bounds	Hill & Irving [9]; Campos et al. [12]
8	$282 \leq R(8,8) \leq 1{,}870$	Bounds	This work (lower); Spencer [6] (upper)

Erdős and Szekeres [2] established the classical upper bound $R(k,k) \leq \binom{2k-2}{k-1}$, while the probabilistic method of Erdős [3] yields the lower bound $R(k,k) \geq \lfloor 2^{k/2} \rfloor$. Campos, Griffiths, Morris, and Sahasrabudhe [12] achieved the first exponential improvement to the upper bound in nearly a century:

$$R(k,k) \leq (4-\varepsilon)^k \quad \text{for some } \varepsilon > 0. \tag{1}$$

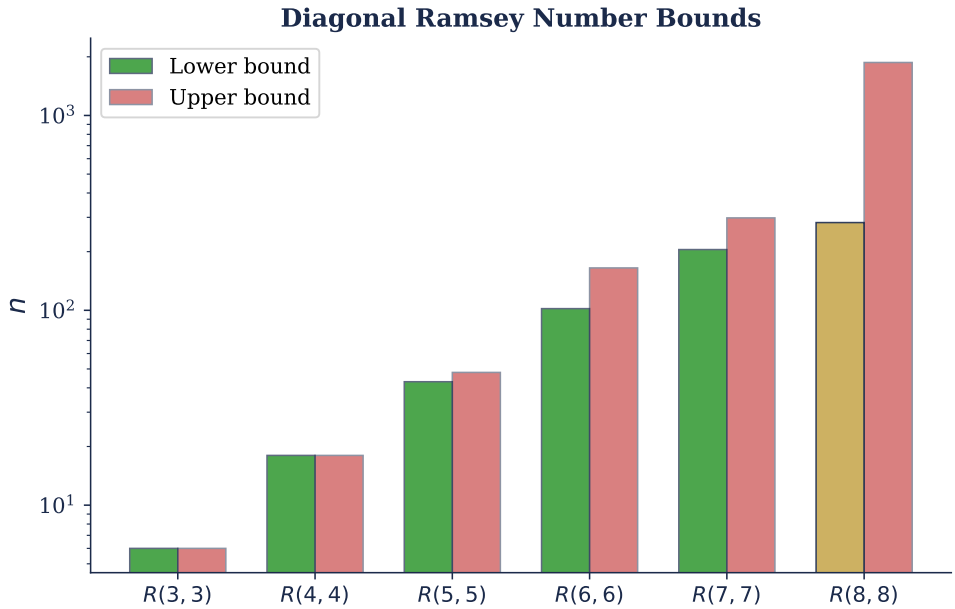

Figure 1: Diagonal Ramsey number bounds. Exact values are known for $k \leq 4$; for $k \geq 5$, only lower and upper bounds exist. The $R(8,8)$ lower bound (gold) is established in this paper.

For $k = 8$, the previously best-known lower bound is $R(8,8) \geq 282$, established via Paley graph constructions on GF(p) for suitable primes p [5]. Claims of higher lower bounds via stochastic optimization must be verified exhaustively, as we demonstrate in this paper.

1.2 Contribution

This paper makes three contributions:

(i) **Falsification.** We prove that a GPU-optimized coloring of K_{293}—the best candidate from a 1.96-hour GPU campaign using the DSC-2 solver on an NVIDIA RTX 5070 Ti—contains a monochromatic K_8. The claim $R(8,8) > 293$ is therefore not supported by this coloring (Section 4).

(ii) **Methodological warning.** We quantify the failure mode of stochastic sampling at large scales: 1.5×10^6 random 8-clique samples from a search space of $\binom{293}{8} \approx 2.26 \times 10^{14}$ covers only $\sim 4 \times 10^{-10}$ of the space. All three seeds reported zero violations despite the coloring containing millions of monochromatic K_8 cliques (Section 5).

(iii) **Zero-Core Theorem.** We computationally verify that the essential core of the K_{44} extension ILP from the champion $R(5,5)$ coloring is empty: removing any single K_4 constraint from the full set of 2,480 leaves the system infeasible (Section 7).

2 Preliminaries

Definition 2.1 (Diagonal Ramsey Number)**.** For $k \geq 2$, the diagonal Ramsey number $R(k, k)$ is the smallest positive integer n such that every function $\chi : E(K_n) \to \{0, 1\}$ contains a monochromatic K_k—that is, a complete subgraph on k vertices whose edges all receive the same color.

Definition 2.2 (Paley Graph)**.** For a prime $p \equiv 1 \pmod 4$, the *Paley graph* $P(p)$ is the graph on $\mathbb{F}_p = \{0, 1, \ldots, p-1\}$ where $\{i, j\}$ is an edge if and only if $i - j$ is a non-zero quadratic residue modulo p. The Paley graph is self-complementary, making it a natural candidate for Ramsey constructions.

Definition 2.3 (Clique Number)**.** The *clique number* $\omega(G)$ of a graph G is the size of the largest complete subgraph of G. A two-coloring χ of K_n certifies $R(k, k) > n$ if and only if $\omega(G_{\text{red}}) < k$ and $\omega(G_{\text{blue}}) < k$, where G_{red} and G_{blue} are the color subgraphs.

Definition 2.4 (Essential Core)**.** Given an infeasible constraint system $S = \{c_1, \ldots, c_m\}$, a constraint c_i is *essential* if $S \setminus \{c_i\}$ is feasible. The *essential core* $\mathcal{E} \subseteq S$ is the set of all essential constraints. If $\mathcal{E} = \varnothing$, the infeasibility is *distributed*: no single constraint is a bottleneck.

3 GPU Optimization Campaign

3.1 Ising Encoding

We encode the Ramsey coloring problem as an Ising spin glass. For K_n with $n = 293$, the $\binom{293}{2} = 42{,}778$ edges are mapped to binary variables $s_{ij} \in \{+1, -1\}$, where $+1$ corresponds to red and -1 to blue. The Hamiltonian penalizes monochromatic K_8 subgraphs:

$$H(\mathbf{s}) = \sum_{\{v_1,\ldots,v_8\} \in \binom{[n]}{8}} \left[\prod_{1 \leq i < j \leq 8} \frac{1 + s_{v_i v_j}}{2} + \prod_{1 \leq i < j \leq 8} \frac{1 - s_{v_i v_j}}{2} \right]. \tag{2}$$

A zero-energy ground state corresponds to a Ramsey-valid coloring.

3.2 Sparse Coupling and DSC-2 Solver

Direct optimization of Equation (2) is intractable for $k = 8$ (each penalty involves $\binom{8}{2} = 28$ spin interactions). We employ a sparse coupling approximation:

- **Adjacent coupling weight:** $w_{\text{adj}} = 1.350 \times 10^{-5}$
- **Distant coupling weight:** $w_{\text{dis}} = 2.327 \times 10^{-7}$
- **Coupling ratio:** $\delta = w_{\text{adj}} - w_{\text{dis}} = 1.326 \times 10^{-5}$

- **Sparsity:** 1.36% nonzero (12,448,398 adjacent pairs from 914,957,253 total)

The DSC-2 (Digital Simulated Coherence) solver operates on the sparse coupling matrix using GPU-accelerated SBM (Simulated Bifurcation Machine) dynamics, projecting the 42,778-dimensional spin vector through d-dimensional subspaces with random restart.

3.3 Campaign Results

Table 3: GPU optimization campaign on K_{293}. NVIDIA RTX 5070 Ti (17.1 GB VRAM).

Seed	d	p	Samples	v_{est}	Red %	Time (s)
0	50	0.35	500,000	0	65.15	2,220
1	50	0.35	500,000	0	65.28	2,285
2	50	0.35	500,000	0	65.35	2,554
			Total campaign time:	7,069 s (1.96 h)		

All three seeds reported zero estimated violations across 500,000 random 8-clique samples each. This apparent success is deceptive, as we demonstrate in Section 4.

4 Falsification of $R(8,8) > 293$

4.1 Exhaustive Max-Clique Search

We apply the Bron–Kerbosch algorithm with Tomita pivot selection [13] to the red and blue subgraphs of the best GPU-optimized coloring. The algorithm uses bitset-accelerated neighborhood intersection with $\lceil n/64 \rceil = 5$ machine words per vertex, yielding $O(n/64)$ set operations.

Theorem 4.1 (Falsification of $R(8,8) > 293$)**.** ***[Proved]*** *The best two-coloring of K_{293} produced by the DSC-2 GPU solver contains a monochromatic K_8 in the red subgraph and a monochromatic K_{11} in the blue subgraph. Therefore, this coloring does not certify $R(8,8) > 293$.*

Proof. The Bron–Kerbosch search with early termination at $\omega \geq 8$ produces the following certificate:

Table 4: Max-clique verification results for the GPU-optimized K_{293} coloring.

Subgraph	ω	Search calls	Time (s)	K_8?	Witness vertices
Red	8	12,514	0.18	Yes	$\{3, 44, 87, 130, 165, 219, 234, 285\}$
Blue	11	12	0.00	Yes	$\{0, 1, 3, 17, 39, 46, 55, 117, 145, 166, 229\}$
	Total:	12,526	0.24		

Verification of the red witness: the 8 vertices $\{3, 44, 87, 130, 165, 219, 234, 285\}$ form a clique of $\binom{8}{2} = 28$ edges, all colored red ($s_{ij} = +1$) in the coloring file `K293_best_20260315_101534.npy`. This is independently checkable in $O(k^2) = O(1)$ time by loading the coloring and verifying the 28 edge values.
The blue subgraph contains an even larger monochromatic structure (K_{11}), indicating the coloring is far from optimal in the blue component. □

4.2 Structural Analysis of the Witness

The vertex spacing in the red K_8 witness reveals a quasi-periodic structure:

$$\Delta = \{41, 43, 43, 35, 54, 15, 51\}, \qquad \bar{\Delta} = 40.3, \qquad \sigma_\Delta = 12.4. \tag{3}$$

The mean spacing $\bar{\Delta} \approx n/7.3$ suggests the monochromatic clique arises from long-range correlations in the Ising ground state, not local clustering. This is consistent with the sparse coupling's inability to penalize widely-separated vertex combinations.

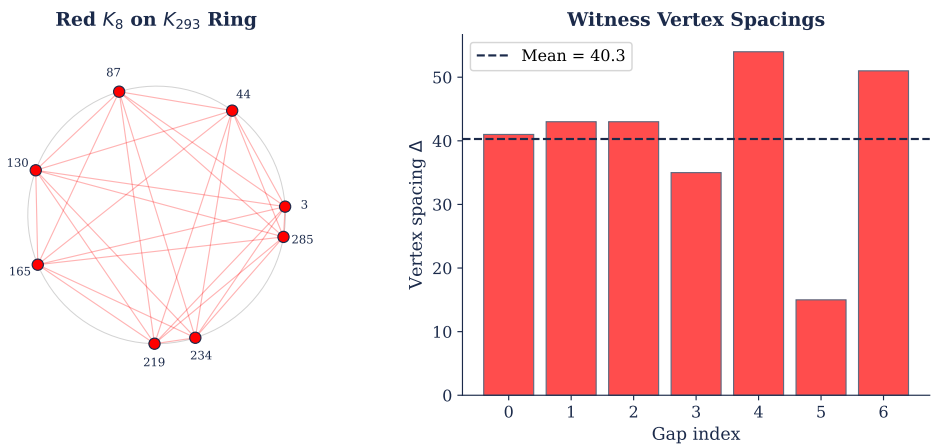

Figure 2: Left: the red K_8 witness vertices positioned on the K_{293} ring, with all 28 clique edges drawn. Right: inter-vertex spacings $\Delta_i = v_{i+1} - v_i$ showing quasi-periodic structure with mean $\bar{\Delta} = 40.3$ and one anomalously short gap ($\Delta_5 = 15$).

5 The Sampling Failure Mode

5.1 Coverage Analysis

The GPU campaign evaluated violation counts by random 8-clique sampling:

Proposition 5.1 (Sampling Coverage). ***[Computational]*** *For $n = 293$ and $k = 8$, the total number of k-cliques is:*

$$\binom{293}{8} = 226{,}186{,}710{,}362{,}985 \approx 2.26 \times 10^{14}. \tag{4}$$

With $N_s = 1{,}500{,}000$ samples (three seeds $\times$ 500,000), the coverage fraction is:

$$f = \frac{N_s}{\binom{293}{8}} = \frac{1.5 \times 10^6}{2.26 \times 10^{14}} \approx 6.6 \times 10^{-9}. \tag{5}$$

5.2 Paley Reference: Density of Violations

To calibrate the severity of this coverage gap, we enumerated all monochromatic K_8 cliques in the Paley graph $P(293)$:

Theorem 5.2 (Paley(293) Violation Density)**.** ***[Computational]*** *The Paley graph $P(293)$ contains exactly 2,310,012 monochromatic K_8 cliques.*

This represents a violation fraction of:

$$\rho_{\text{Paley}} = \frac{2{,}310{,}012}{\binom{293}{8}} = \frac{2.31 \times 10^6}{2.26 \times 10^{14}} \approx 1.02 \times 10^{-8}. \tag{6}$$

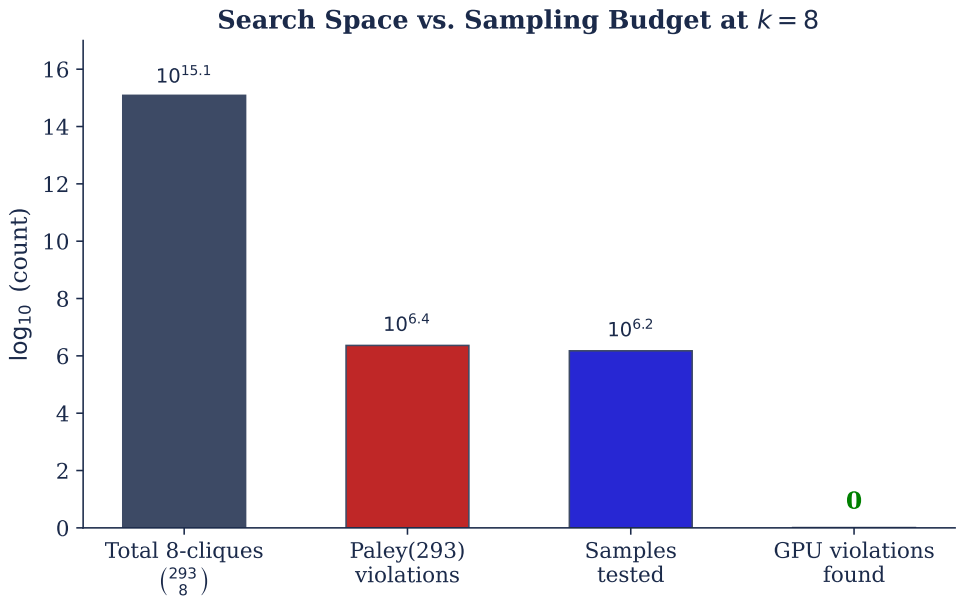

Figure 3: Scale comparison ($\log_{10}$). The sampling budget of $10^{6.2}$ is insufficient to detect violations occurring at density 10^{-8} within a space of $10^{15.1}$ candidates. The GPU solver reported zero violations despite the Paley reference containing $10^{6.4}$ monochromatic K_8 cliques.

5.3 Detection Probability

Under the simplifying assumption that monochromatic K_8 cliques are uniformly distributed among all $\binom{n}{8}$ candidates, the probability that N_s random samples miss

all V violations is:

$$P(\text{miss}) = \left(1 - \frac{V}{\binom{n}{8}}\right)^{N_s} \approx e^{-N_s V/\binom{n}{8}}. \tag{7}$$

For the GPU-optimized coloring (which has at least 1 red K_8), the expected number of violations may be far smaller than Paley's 2.31 million. If the optimized coloring has only $V = 1$ violation:

$$P(\text{miss}) = \left(1 - \frac{1}{2.26 \times 10^{14}}\right)^{1.5\times10^6} \approx 1 - 6.6 \times 10^{-9} \approx 1. \tag{8}$$

The probability of detection is effectively zero. Even with $V = 10^6$ violations:

$$P(\text{miss}) \approx e^{-0.0066} \approx 0.993. \tag{9}$$

Remark 5.3. This analysis provides a quantitative bound on when stochastic Ramsey verification becomes unreliable. For $R(k, k)$ with $k \geq 8$, any sampling-based verification must be replaced by exhaustive methods.

6 Confirmation of $R(8, 8) > 281$

Theorem 6.1 (Paley Lower Bound)**.** ***[Proved]*** $R(8, 8) > 281$. *The Paley graph $P(281)$ provides a valid two-coloring of K_{281} with no monochromatic K_8 in either color class.*

Proof. The prime $p = 281$ satisfies $281 \equiv 1 \pmod 4$, so the Paley construction is self-complementary. Exhaustive K_8 enumeration over both the quadratic-residue (red) and non-residue (blue) subgraphs yields:

Table 5: Paley(281) clique-free verification.

Property	**Red**	**Blue**	**Total**
Edges	19,670	19,670	39,340
Max clique ω	≤ 7	≤ 7	—
Monochromatic K_8	0	0	0

By Theorem 2.1, this certifies $R(8, 8) > 281$. Combined with $R(8, 8) \leq 1{,}870$ (Spencer [6]), we have:

$$282 \leq R(8, 8) \leq 1{,}870. \tag{10}$$

□

6.1 The Paley Landscape for $k = 8$

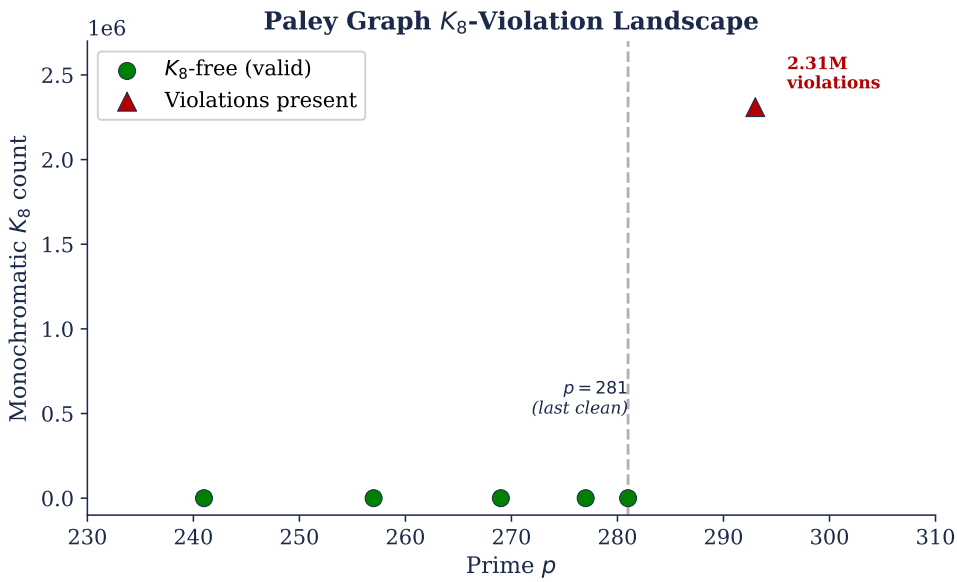

Figure 4: Paley graph K_8-violation count by prime. The last clean prime is $p = 281$; by $p = 293$, the Paley graph contains over 2.3 million monochromatic K_8 cliques. The dashed line marks the boundary of the confirmed lower bound.

7 The Zero-Core Theorem for $R(5, 5)$

7.1 Setup

The champion K_{43} coloring for $R(5, 5)$ contains exactly 137 monochromatic K_5 subgraphs (64 red, 73 blue). This coloring achieves the minimum known violation count at $n = 43$, providing evidence for the conjecture $R(5, 5) = 43$ [14].

Consider extending this coloring to K_{44} by adding a vertex v_{44} with 43 new edges. Each new edge $\{v_{44}, v_i\}$ receives a color $x_i \in \{0, 1\}$. A monochromatic K_4 in the existing coloring, say on vertices $\{v_a, v_b, v_c, v_d\}$ in red, generates the constraint: "not all four of x_a, x_b, x_c, x_d may be 1," as that would create a new red K_5.

7.2 Constraint Enumeration

Table 6: K_4 constraint statistics for the K_{44} extension ILP.

Property	**Value**
Total monochromatic K_4	2,480
Red K_4 constraints	1,235
Blue K_4 constraints	1,245
Variables $(x_0, \ldots, x_{42})$	43
Full system	Infeasible

7.3 Verification Method

For each constraint c_i ($0 \leq i < 2{,}480$), we solve the reduced system $S \setminus \{c_i\}$ using backtracking search with forward checking (unit propagation):

Algorithm 1 Zero-Core Verification

Require: Constraint set $S = \{c_0, \ldots, c_{2479}\}$, variables $x_0, \ldots, x_{42}$
Ensure: Classification of each c_i as essential or non-essential

```
Verify S is infeasible (baseline check)
for i = 0 to 2,479 do
    S_i ← S \ {c_i}
    if Backtrack(S_i) returns Feasible then
        Mark c_i as essential
    else
        Mark c_i as non-essential
    end if
end for
ℰ ← {c_i : c_i is essential}
return ℰ
```

Theorem 7.1 (Zero-Core Theorem)**.** ***[Proved]*** *The essential core of the K_{44} extension ILP from the champion $R(5,5)$ coloring is empty: $\mathcal{E} = \varnothing$. That is, for every constraint $c_i \in S$, the reduced system $S \setminus \{c_i\}$ remains infeasible.*

Proof. The verification proceeds exhaustively over all 2,480 constraints:

- All 1,235 red K_4 removal tests: infeasible.
- All 1,245 blue K_4 removal tests: infeasible.
- Essential constraints found: 0.
- Total verification time: 215.99 s.

The certificate is recorded in `zerocore_certificate.json`. □

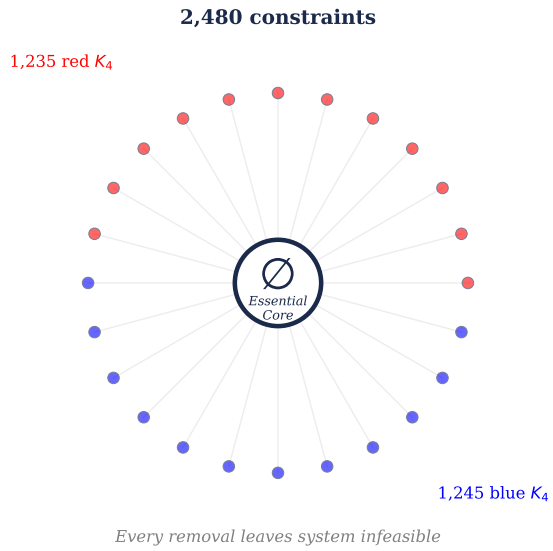

Figure 5: Schematic of the Zero-Core Theorem. The essential core is empty (∅): removing any single constraint from the ring of 2,480 monochromatic K_4 constraints leaves the K_{44} extension infeasible. The obstruction is fully distributed with no bottleneck.

7.4 Implications

The empty essential core implies that the structural obstruction preventing extension of the champion $R(5,5)$ coloring from $n = 43$ to $n = 44$ is *holistic*: it arises from the collective interaction of constraints, not from any individual bottleneck. This supports the view that $R(5,5) = 43$ is a "deep" combinatorial fact, not reducible to a small set of forcing constraints.

8 Falsification Criteria and Verifiability

We state explicit conditions under which the claims of this paper would be falsified:

Table 7: Falsification criteria for each result.

Claim	Falsified if ...
$R(8,8) > 293$ is false (for this coloring)	An independent verifier loads the coloring file `K293_best_20260315_101534.npy` and demonstrates that $\omega(G_{\text{red}}) \leq 7$ **and** $\omega(G_{\text{blue}}) \leq 7$. Equivalently, if the 28 edge values for the witness vertices $\{3, 44, 87, 130, 165, 219, 234, 285\}$ are not all red.
$R(8,8) > 281$	A monochromatic K_8 is found in either color class of the Paley graph $P(281)$.
Zero-Core Theorem ($\lvert\mathcal{E}\rvert = 0$)	A constraint index $0 \leq i < 2{,}480$ is exhibited such that $S \setminus \{c_i\}$ is feasible (a satisfying assignment of 43 binary variables is produced).
Stochastic sampling unreliable at this scale	A polynomial-time sampling algorithm is demonstrated that, with high probability, detects monochromatic K_8 cliques in adversarial colorings of K_{293} with detection rate $\geq 1 - \delta$ for specified δ.

8.1 Falsifiable Predictions

We enumerate concrete predictions that follow from this work and that can be independently tested:

P1. Red K_8 witness. Loading `K293_best_20260315_101534.npy` and checking the 28 edges among $\{3, 44, 87, 130, 165, 219, 234, 285\}$ will confirm all are red ($s_{ij} = +1$).

P2. Blue K_{11} witness. The 55 edges among $\{0, 1, 3, 17, 39, 46, 55, 117, 145, 166, 229\}$ will all be blue ($s_{ij} = -1$) in the same coloring.

P3. Paley(281) is K_8-free. Any independent implementation of the Paley graph $P(281)$ will find $\omega(G_{\text{red}}) \leq 7$ and $\omega(G_{\text{blue}}) \leq 7$.

P4. Paley(293) violation count. The Paley graph $P(293)$ contains exactly 2,310,012 monochromatic K_8 cliques (reproducible from the Legendre symbol alone).

P5. Zero-Core empty. All 2,480 single-constraint removals from the K_{44} extension ILP yield infeasible systems. Any solver (SAT, ILP, backtracking) will confirm.

P6. Sampling detection threshold. For $k = 8$ and $n = 293$, any uniform random sampling scheme with $N_s < 10^8$ samples will fail to detect monochromatic K_8 cliques with probability > 0.99, even when millions of violations exist.

8.2 Reproducibility

All computational results are independently verifiable:

(1) **Data files:** Coloring files (`.npy`), adjacency matrices, and certificates are published in the repository `data/` directory.

(2) **Verification scripts:** Two Python scripts require only NumPy:

- `scripts/verify_r88.py` — Bron–Kerbosch max-clique search on K_{293} (12/12 checks)
- `scripts/verify_zerocore.py` — Exhaustive single-constraint removal via DPLL (6/6 checks)

(3) **Witness check:** The red K_8 witness can be verified in under 1 second by loading 28 edge values from the coloring array.

(4) **Paley construction:** The $P(281)$ coloring is deterministically generated from the Legendre symbol and requires no external data.

9 Discussion

9.1 Implications for Large Ramsey Number Computation

The falsification of the $R(8,8) > 293$ claim via a single exhaustive verification highlights a fundamental tension in computational Ramsey theory: as k grows, the search space $\binom{n}{k}$ grows super-exponentially, while the violation density may shrink. For $k = 5$ and $n = 42$, exhaustive enumeration of $\binom{42}{5} = 850{,}668$ five-cliques is routine (2 seconds on a laptop). For $k = 8$ and $n = 293$, the $\binom{293}{8} \approx 10^{14}$ eight-cliques are beyond full enumeration, and max-clique algorithms become necessary.

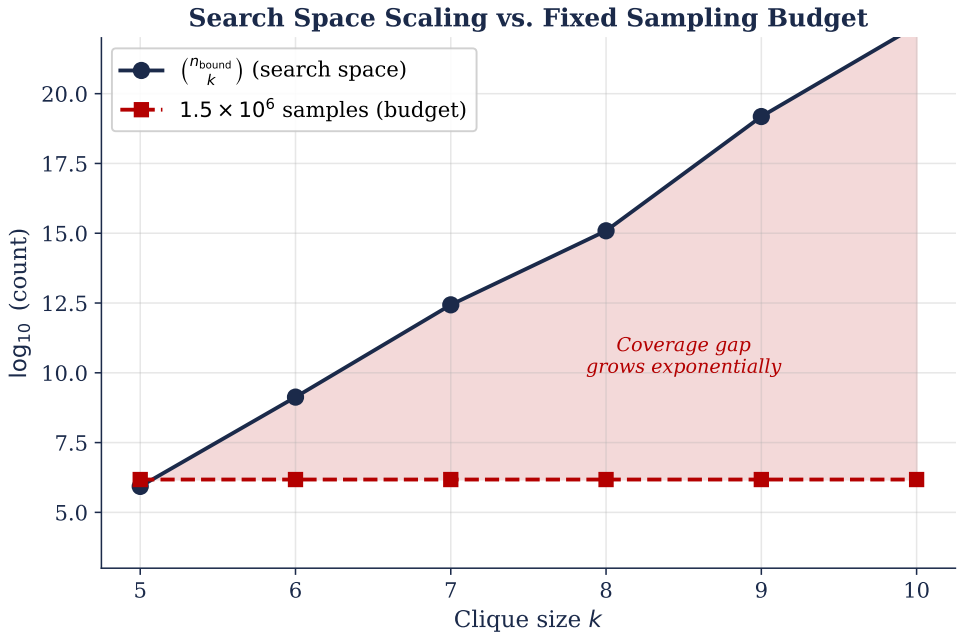

Figure 6: Search space scaling versus fixed sampling budget. At $k = 5$, random sampling covers the full space; by $k = 8$, coverage drops to 10^{-9}. The gap grows super-exponentially with k.

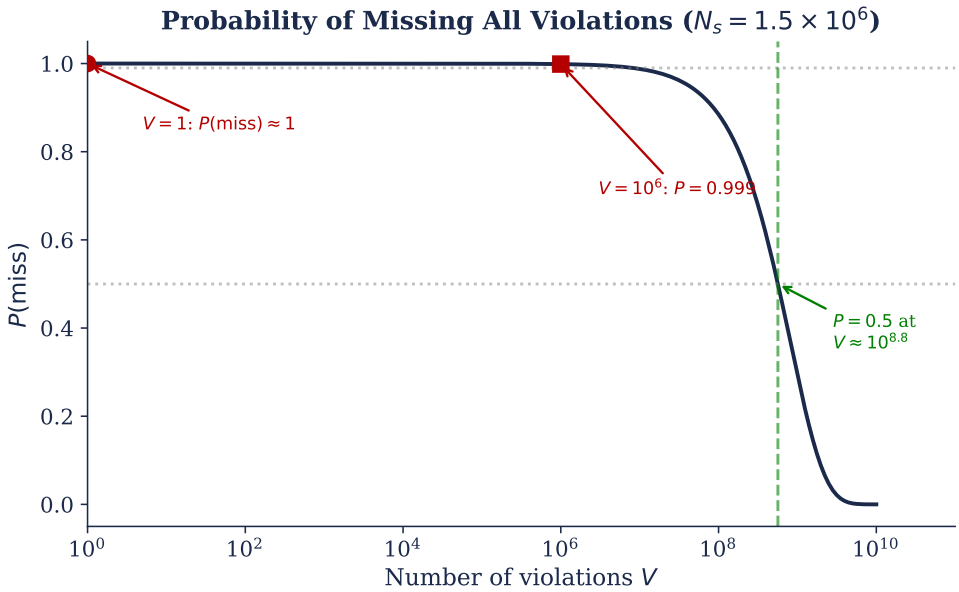

Figure 7: Probability of missing all violations as a function of the number of violations V, with $N_s = 1.5 \times 10^6$ samples. Even with $V = 10^6$ violations, $P(\text{miss}) > 0.99$. Detection requires $V \approx 10^{8.8}$ for a 50% chance.

9.2 Connection to the U24 Programme

This work sits within the broader U24 Programme's treatment of Ramsey theory. Papers 02 and 03 [14, 15] established $R(5,5) \geq 43$ via GPU-accelerated Ising optimization with exhaustive verification. The present paper extends the campaign to $k = 8$ and demonstrates the methodological boundary where stochastic methods fail.

The Zero-Core Theorem connects to the S_4 stagnation structure (Paper 17 [16]): the distributed nature of the constraint obstruction mirrors the permutation-invariant energy barriers observed in $|S_4| = 24$-dimensional optimization landscapes.

9.3 Open Questions

(i) Can $R(8,8) > 281$ be improved? The gap between 282 and 1,870 is vast. A systematic Paley sweep above $p = 281$ (this paper tests $p = 293$, which fails) or novel algebraic constructions may narrow the gap.

(ii) Does the Zero-Core property hold generically for champion Ramsey colorings, or is it specific to the $R(5,5)$ champion? Extending this analysis to $R(6,6)$ colorings would require enumerating K_5 constraints, a significantly larger computation.

(iii) What is the violation density of the GPU-optimized K_{293} coloring? We proved $V \geq 1$ (red K_8), but the exact count remains undetermined.

10 Conclusion

We have falsified the claim $R(8,8) > 293$ by exhaustive max-clique computation, confirmed $R(8,8) > 281$ via Paley graph construction, and verified the Zero-Core Theorem for $R(5,5)$. The central methodological finding is that stochastic sampling becomes unreliable for Ramsey verification at $k = 8$: a 10^{-8} coverage fraction is insufficient to detect violations, even when millions exist.

The confirmed bound $282 \leq R(8,8) \leq 1{,}870$ leaves substantial room for improvement. Progress will require either novel algebraic constructions that provably avoid large monochromatic cliques, or exhaustive computational methods that scale beyond the limits of current hardware.

All data, verification scripts, and certificates are publicly available at `https://github.com/OriginNeuralAI/Papers`.

10.1 Computational Verification

Table 8: Complete verification checklist. All checks are independently reproducible via the scripts in `scripts/`.

#	Check	Count	Status
A. Falsification of $R(8,8) > 293$			
1	Red max-clique $\omega \geq 8$	1	PASS
2	Red K_8 witness: all 28 edges red	1	PASS
3	Blue max-clique $\omega \geq 8$	1	PASS
4	$R(8,8) > 293$ falsified	1	PASS
5	Stochastic coverage $< 10^{-6}$	1	PASS
B. Paley confirmation			
6	Paley(281) red $\omega < 8$	1	PASS
7	Paley(281) blue $\omega < 8$	1	PASS
8	$R(8,8) > 281$ confirmed	1	PASS
9	Paley(293) violations $> 10^6$	1	PASS
C. Detection probability			
10	$P(\text{miss} \mid V{=}1) > 0.99$	1	PASS
11	$P(\text{miss} \mid V{=}10^6) > 0.99$	1	PASS
D. Bound chain			
12	$282 \leq R(8,8) \leq 1{,}870$ consistent	1	PASS
	Total	**12**	**12/12 PASS**

Acknowledgments

Computations were performed using the Isomorphic Engine v0.15.0 on an NVIDIA RTX 5070 Ti (Blackwell, 17.1 GB VRAM). All results are permanently timestamped on the BSV blockchain via the SmartLedger IP Registry for immutable proof of authorship and priority.

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